

Phase 0: Building the Dataset

Project: मोडी ते मराठी (modi-to-Marathi) — Modi script → Marathi (Devanagari)

Repository: <https://github.com/lgtkgvtv/ai4-transliteration-modi-to-marathi> **From:** Sachin, Project Maintainer **To:** Project contributors and faculty advisors **Date:** [Date]
Re: Our first real piece of work — and why it isn't the model

The core message

Team — it's tempting to think of this as an "AI model" project, but I want to set expectations clearly: **our first and most important sub-project is building a high-quality, labeled dataset of Modi documents paired with verified Marathi (Devanagari) transliterations.** Every successful project I've studied in this area succeeded or failed on the strength of its dataset and its expert-review process, not its choice of model. The model is the comparatively easy part.

So Phase 0 is about data. Below is how I propose we approach it. Please challenge anything that doesn't fit our reality.

1. What a "labeled example" means for us

For Modi → Marathi, a labeled example is:

an image of Modi text → its correct, expert-verified Marathi transliteration in Devanagari.

We can't build anything until we agree on the exact unit (a full page? a few lines? a single line?) and a consistent way to produce these pairs. The reference project segmented each document into parts of **3–4 lines** before transliterating — a reasonable starting convention for us to discuss.

2. Do this first: a small, gold-standard test set

Before generating any volume of data, we need a **small set of examples verified to be correct beyond doubt** — on the order of a few hundred. This is non-negotiable, for one reason:

Without a trustworthy test set, we cannot tell whether anything we later build is actually working. Every accuracy number we ever report depends on it.

This test set should be: - transliterated by our **most qualified Modi experts**, - **cross-checked** by a second expert, - **locked and never used for training** — kept aside purely for measurement, - ideally **spanning the eras and styles** we expect to encounter, so it's representative.

If we do nothing else in month one, we should produce this.

3. How we get from raw Modi images to labeled data efficiently

We won't have unlimited expert time, so the strategy is to **minimize from-scratch labeling** and convert expert effort into correcting drafts wherever possible.

Strategy	What it is	Viability for Modi → Marathi
Expert gold labeling	Experts produce verified pairs from scratch	Required for the test set and a small training seed
Synthetic generation	Render Modi images from known Marathi/Devanagari text using Modi fonts	Strong option for us — see section 4
Reuse existing datasets	Build on already-published Modi datasets	Available — see section 5
Bootstrap-and-correct	A rough first model drafts labels; experts only fix errors	High value once we have any working model
Active learning	Prioritize labeling the examples the model is least sure about	To stretch limited expert hours

Strategy	What it is	Viability for Modi → Marathi
Correction flywheel	Model drafts → expert corrects → corrections become new training data	Ongoing, compounds over time

The principle behind all of these: **correcting a draft is far cheaper than writing a label from scratch.**

4. Synthetic data is genuinely viable for us

Because open **Modi fonts exist**, we can render Modi-script images from any Marathi (Devanagari) text we have — generating large amounts of training data cheaply. The reference project did exactly this using:

- **MarathiCursive** (an open Modi/Marathi cursive font)
- **Noto Sans Modi** (Google's open Modi font)

Synthetic data won't capture the messiness of real faded manuscripts, so it can't replace real data — but it's an excellent way to **bulk up training cheaply** and pre-train before fine-tuning on our real, expert-verified examples. **Worth piloting early.**

5. Existing Modi datasets we should investigate for reuse

We don't have to start completely empty. These have been published (mostly via IEEE DataPort / Hugging Face) and are worth evaluating — with the caveat that **most are character-level, not full-document transliteration**, so they help recognition more than end-to-end transliteration:

Dataset	Type
historyHulk/MoDeTrans	Full Modi documents → Devanagari (closest to our task)
historyHulk/SynthMoDe	Synthetic Modi → Devanagari
Modi-HHDoc	~3,350 Modi documents

Dataset	Type
MODI-HChar	Individual Modi characters
Handwritten Modi Lipi Barakhadi	Character/syllable set
Handwritten Modi Characters	Individual characters

Advisors — please tell me which of these you trust, and whether their licenses allow our intended use.

6. Quality standards we should agree on now

To keep the dataset trustworthy and publishable later, I propose we adopt these from day one:

- **Written transliteration guidelines** so every contributor is consistent — explicitly covering the known confusable Modi characters (भ/म, क/फ, ट/ठ/ढ, न/ण), vowel signs, and the **anusvāra** dot, plus how to mark uncertain readings.
 - **A second-reviewer pass** on a defined percentage of examples.
 - **Strict separation** of training, validation, and test data — with a manual check that no item appears in more than one (the reference team used 10 annotators for exactly this).
 - **Provenance tracking** — for every item, record its source archive and who verified it (also important for licensing if we publish).
 - **A consistent, documented preprocessing recipe** for images (grayscale → denoise → adaptive threshold → deskew → consistent sizing), applied identically to every item.
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7. Proposed roles

Role	Who	Responsibility
Source contributors	[names]	Provide Modi material in an agreed format, with provenance
Expert transliterators	[names]	Produce and verify the gold and seed labels

Role	Who	Responsibility
Reviewers	[names]	Second-pass cross-checking
Engineering	Sachin + [names]	Preprocessing, tooling, the correction interface, split management
Coordination	Sachin	Guidelines, schedule, quality tracking

Please correct these — I've guessed at the structure.

8. Minimum viable targets (to refine with your input)

We don't need a massive corpus to begin. A realistic v1 footing: - a **few hundred** locked, expert-verified test examples (representative across eras), - a **seed set** of verified real training examples, - a **larger synthetic set** (rendered with Modi fonts) to bulk up training cheaply, - optionally, **vettred reuse** of an existing dataset from section 5.

Exact numbers depend on the scoping answers (expert availability, source condition, era variation). I'll propose firm targets once those come back.

9. Risks I want us to manage from the start

Risk	Why it hurts	How we mitigate
Too few Modi experts	Labeling becomes the project bottleneck	Reserve experts for the test set + corrections, not bulk labeling
Inconsistent labels	Quietly destroys accuracy and trust	Shared written guidelines + second reviewer
Data leakage	Makes our results look better than they are	Manual split verification, provenance tracking
Over-reliance on synthetic data	Looks good in tests, fails on real manuscripts	Always measure on real held-out documents
Skipping the test set	We can't measure progress at all	Build it first, before anything else

Risk	Why it hurts	How we mitigate
Over-building the model early	Wasted effort before data is ready	Baseline with existing tools before training our own

10. What I need from each of you to start

- **Contributors:** confirm the format and provenance details, and start assembling a first small batch.
- **Experts/advisors:** help me draft the transliteration guidelines and define what "verified" means for Modi, and tell me which existing datasets in section 5 we can trust and legally reuse.
- **Everyone:** agree on the definition of a "labeled example" (section 1) so we're all building the same thing.

If we get the gold test set, the guidelines, and a synthetic-data pilot right in the first stretch, everything afterward gets much easier. Happy to walk any of this through live.